

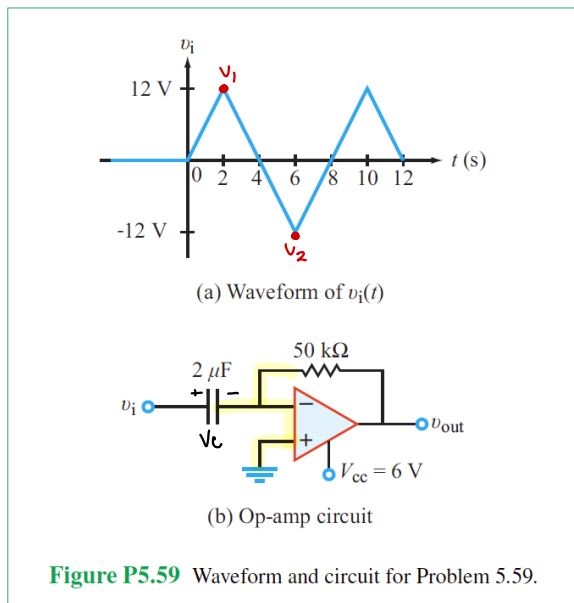
Name: Anyia Scott (please print)
1599670

ECE 2202 - HW4

R_c differentiation

1. Textbook #5.59 (Solution omitted)

5.59 The input-voltage waveform shown in Fig. P5.59(a) is applied to the circuit in Fig. P5.59(b). Determine and plot the corresponding $v_{out}(t)$.



$$V_c = V_i$$

$$-i_i - i_c = 0$$

$$i_i = \frac{V_{out}}{50[k\Omega]}$$

$$i_c = \frac{dV_i}{dt} \cdot C$$

$$\frac{V_{out}}{50[k\Omega]} = \frac{dV_i}{dt} \cdot C (50[k\Omega])$$

$$V_{out} = 50[k\Omega] \cdot C \frac{dV_i}{dt}$$

$$C = 2e-6$$

$$V_i = 6t \rightarrow 6t \frac{d}{dt} = 6$$

$$V_{out} = 50[k\Omega] (2[\mu F]) \cdot 6 = -0.6[V]$$

$$2 \leq t \leq 6$$

$$\frac{V_2 - V_1}{t_2 - t_1} \rightarrow \frac{(-12 - 12)[V]}{6[s] - 2[s]} = -6$$

$$V_{out} = 0.6[V]$$

$$6 \leq t \leq 10 \rightarrow -0.6[V]$$

$$t > 10 \rightarrow 0.6[V]$$

this pattern of this repeats when you plug for V_{out} with time constraints.

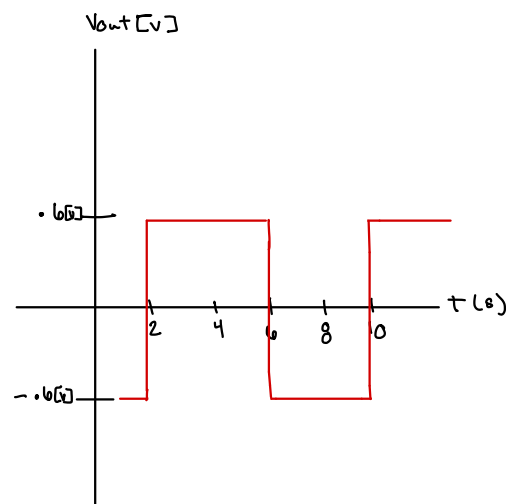
V_i :

$$0 \leq t \leq 2 \rightarrow 6t$$

$$2 \leq t \leq 6 \rightarrow -6t$$

$$6 \leq t \leq 10 \rightarrow 6t$$

$$t \geq 10 \rightarrow -6t$$



2. Textbook #5.60 (Solution in textbook)

*5.60 Relate v_{out} to v_i in the circuit of Fig. P5.60.

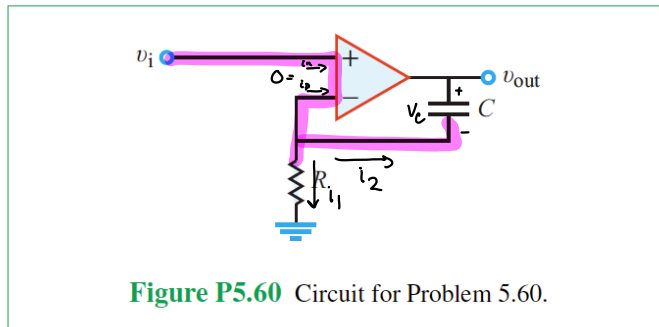


Figure P5.60 Circuit for Problem 5.60.

RC Integrator

$$i_1 + i_2 = 0 \text{ [A]} \quad i_1 = \frac{v_i}{R_1} \quad ; \quad i_2 = C \left(\frac{dv_{out} - v_i}{dt} \right)$$

$$\frac{v_i}{RC} = \cancel{\left(\frac{dv_{out} - v_i}{dt} \right)} = 0 \rightarrow \frac{v_i}{RC} = \frac{d(v_{out} - v_i)}{dt} = 0$$

↓

$$\frac{1}{RC} \int_0^t v_i dt = \int_0^t \frac{dv_{out}}{dt} dt - \int_0^t \frac{dv_i}{dt} dt$$

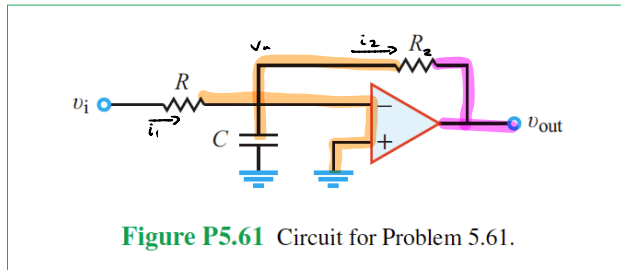
$\hookrightarrow v_{out}(t) - v_{out}(0)$

$$\frac{1}{RC} \int_0^t v_i dt + \int_0^t dv_i = v_{out}(t) - v_{out}(0)$$

$$v_{out}(t) = \frac{1}{RC} \int_0^t v_i dt + \int_0^t dv_i + v_{out}(0)$$

3. Textbook #5.61 (Solution omitted)

5.61 Develop the relationship between the output voltage v_{out} and the input voltage v_i for the circuit in **Fig. P5.61**.



$$v_n = 0 \text{ [v]}$$

$$\frac{v_i - v_n}{R} \rightarrow \frac{v_i}{R}$$

$$I_c = C \cdot \frac{d}{dt}(v_n) = C \frac{d}{dt}(0) = C \frac{d}{dt}(v_i)$$

$$I_2 = \frac{v_n - v_{out}}{R} = \frac{-v_{out}}{R}$$

$$\frac{v_i}{R} - 0 - \frac{v_{out}}{R} = 0 \rightarrow \frac{v_i}{R} = \frac{v_{out}}{R} \rightarrow v_i - v_{out}$$

$$I_1 = I_2 + I_c$$

$$\frac{v_i}{R} = \frac{-v_{out}}{R} + C \frac{dv_i}{dt}$$

$$\frac{v_{out}}{R} = \frac{-v_i}{R} + C \frac{dv_i}{dt} \rightarrow v_{out} = -v_i + RC \frac{dv_i}{dt}$$

4. Textbook #5.64 (Solution omitted)

5.64 Determine $v_{out}(t)$ in the circuit of Fig. P5.64 for $t \geq 0$.

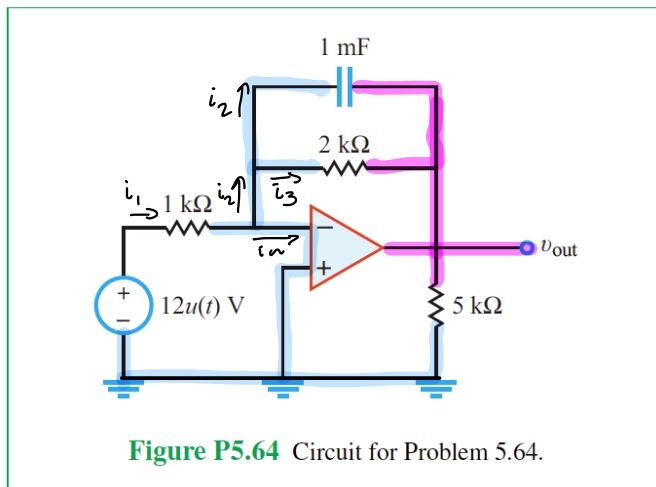


Figure P5.64 Circuit for Problem 5.64.

$$i_1 + i_2 + i_3 + i_4 = 0$$

$$i_1 = \frac{V_1}{R_1}$$

$$i_2 = C \frac{dv_{in} - v_{out}}{dt}$$

$$i_3 = \frac{v_{in} - v_{out}}{R_2}$$

Integrator - $0[V], t < 0(s)$
 $12[V], t \geq 0(s)$

$$\tau = \frac{1}{10^{-3}(2 \times 2)} = 0.5(s)$$

$$RC = 2[k\Omega](1[mF]) = 2[s]$$

$$x(t) = x(0)e^{-at} + \frac{b}{a}(1 - e^{-at}) \leftarrow \text{this is from the general equation of } \frac{dx}{dt} + ax = b$$

$$v_{out} = -\frac{V_s R_2}{R_1} (1 - e^{-at})$$

$$\frac{-12[V](2[k\Omega])}{1[k\Omega]} = -24[V](1 - e^{-0.5t})$$

$$v_{out} \ t \geq 0 = -24[V](1 - e^{-0.5[s](t \geq 0)}) u(t)$$

dis regard this $\downarrow$. This method is from circuits 1

$$V_{out}(\infty) = \left(\frac{-R_f}{R_i}\right) V_i \Rightarrow \left(\frac{-2k\Omega}{1k\Omega}\right) 12[V] = -24[V]$$

$$V_{out}(\infty) + (V_{out}(0) - V_{out}(\infty))e^{-t/\tau}$$

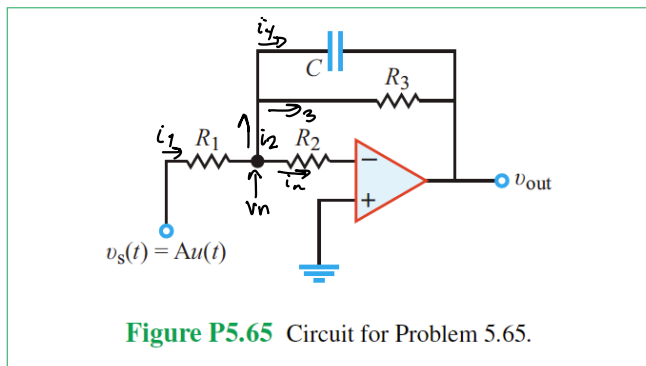
$$-24[V] + 0 - 24[V]e^{-t/2[s]}$$

5. Textbook #5.65 (Solution in textbook for part b)

5.65 In the circuit of Fig. P5.65:

(a) Derive an expression for $v_{out}(t)$ for $t \geq 0$ in terms of R_1 , R_2 , R_3 , C , and A .

*(b) Evaluate the expression for $R_1 = 1 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$, $C = 0.25 \text{ mF}$, and $A = 12 \text{ V}$.



$t \geq 0$

$$i_1 = \frac{v_n - v_s}{R_1} + \frac{v_n}{R_2} = 0$$

$$i_2 = C \frac{d}{dt}(v_n - v_{out}) + \frac{v_n - v_{out}}{R_3} = 0$$

$$\frac{v_n - v_s}{R_1} + \frac{v_n}{R_2} + \left(\frac{d}{dt}(v_n - v_{out}) + \frac{v_n - v_{out}}{R_3} \right) = 0$$

$$\frac{-v_s}{R_1} - \frac{d}{dt}(v_{out} - \frac{v_{out}}{R_3}) = 0$$

$$\frac{dv_{out}}{dt} + a v_{out} = b$$

$$v_{out} = \frac{-v_s R_3}{R_1} (1 - e^{-at})$$

$$v_{out} = \frac{-v_s R_3}{R_1} (1 - e^{-at}) \quad t \geq 0$$

segments:

$t \geq 0$

b)

$$x(t) = x(0)e^{-at} + \frac{a}{b}(1 - e^{-at})$$

$$v_{out}(0)e + \frac{CR_3}{CR_1}(1 - e^{-t/\tau})$$

$$v_{out}(0)e^{-\frac{t}{\tau}} + \frac{-v_s R_3}{R_1}(1 - e^{-t/\tau})$$

$$v_{out} = -v_{c1}$$

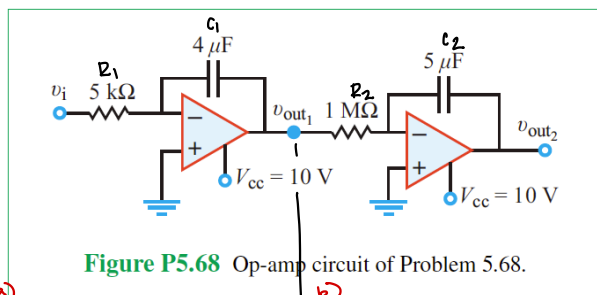
$$v_{out}(t) = \frac{-v_s R_3}{R_1} (1 - e^{-t/\tau})$$

$$\frac{-12[V](2[k\Omega])}{1[k\Omega]} = -24[V](1 - e^{-2t})[V]$$

6. Textbook #5.68 (Solution in textbook for part a)

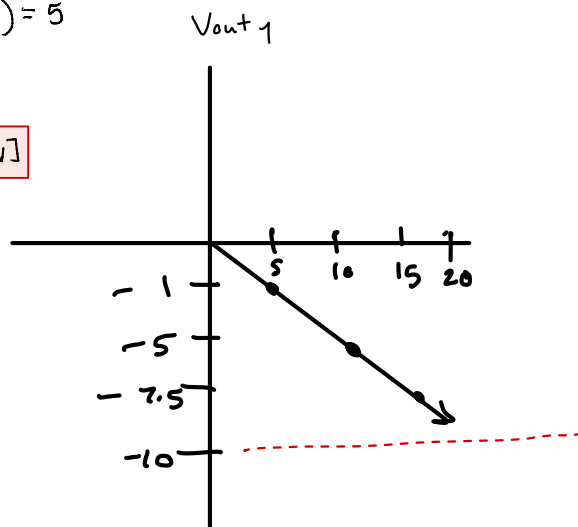
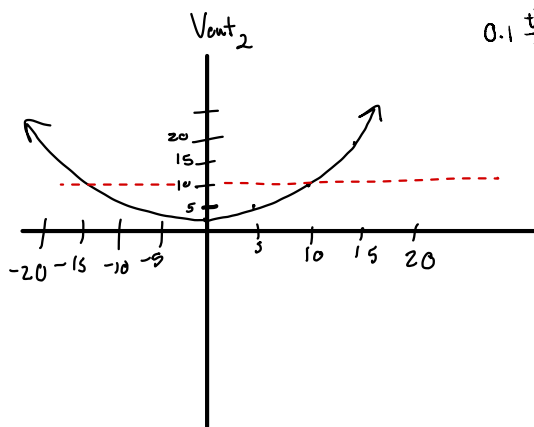
5.68 The two-stage op-amp circuit in **Fig. P5.68** is driven by an input step voltage given by $v_i(t) = 10u(t)$ mV. If $V_{cc} = 10$ V for both op amps and the two capacitors had no charge prior to $t = 0$, determine and plot:

- ***(a)** $v_{out1}(t)$ for $t \geq 0$;
- (b)** $v_{out2}(t)$ for $t \geq 0$.



$$\begin{aligned} & \text{a) } v_{out1} \quad t > 0 \quad \frac{-1}{RC} = -60[s^{-1}] \\ & v_{out1} = \frac{-1}{RC} \int_0^t u(t) dt \\ & \frac{-1}{4e-6(5e3)} \int_0^t 10 dt \\ & -60 \cdot 10t \Big|_0^t \rightarrow \\ & \boxed{-60 \int_0^t 10 dt \rightarrow v_{out1} = -600t [mV]} \end{aligned}$$

$$\begin{aligned} & \text{b) } t > 0 \quad v_{out2} \\ & C_1 = 1 + \left(\frac{R_2}{R_1}\right) = 1 + \frac{1[M\Omega]}{5[k\Omega]} = 200[-] \\ & v_{out2} = \frac{1}{R_2 C_2} \int_0^t v_{out1}(t) dt \\ & R_2 C_2 = (1e6[s^{-1}]) (5[\mu F]) = 5 \\ & \frac{-1}{5} \int_0^t (-0.6t) dt \\ & 0.1 \frac{t^2}{2} = \boxed{0.05t^2 [V]} \quad t \geq 0 \end{aligned}$$



Solutions

1. Solution omitted ✓

2. Solution in textbook ✓

3. Solution omitted ✓

4. Solution omitted ✓

5. b) $v_{\text{out}}(t) = -24(1 - e^{-2t})$ $t \geq 0$. ✓

6. a) $v_{\text{out}_1}(t) = -50 \int_0^t 10 \times 10^{-3} dt = -0.5t$ (V), ✓ for $t \geq 0$.